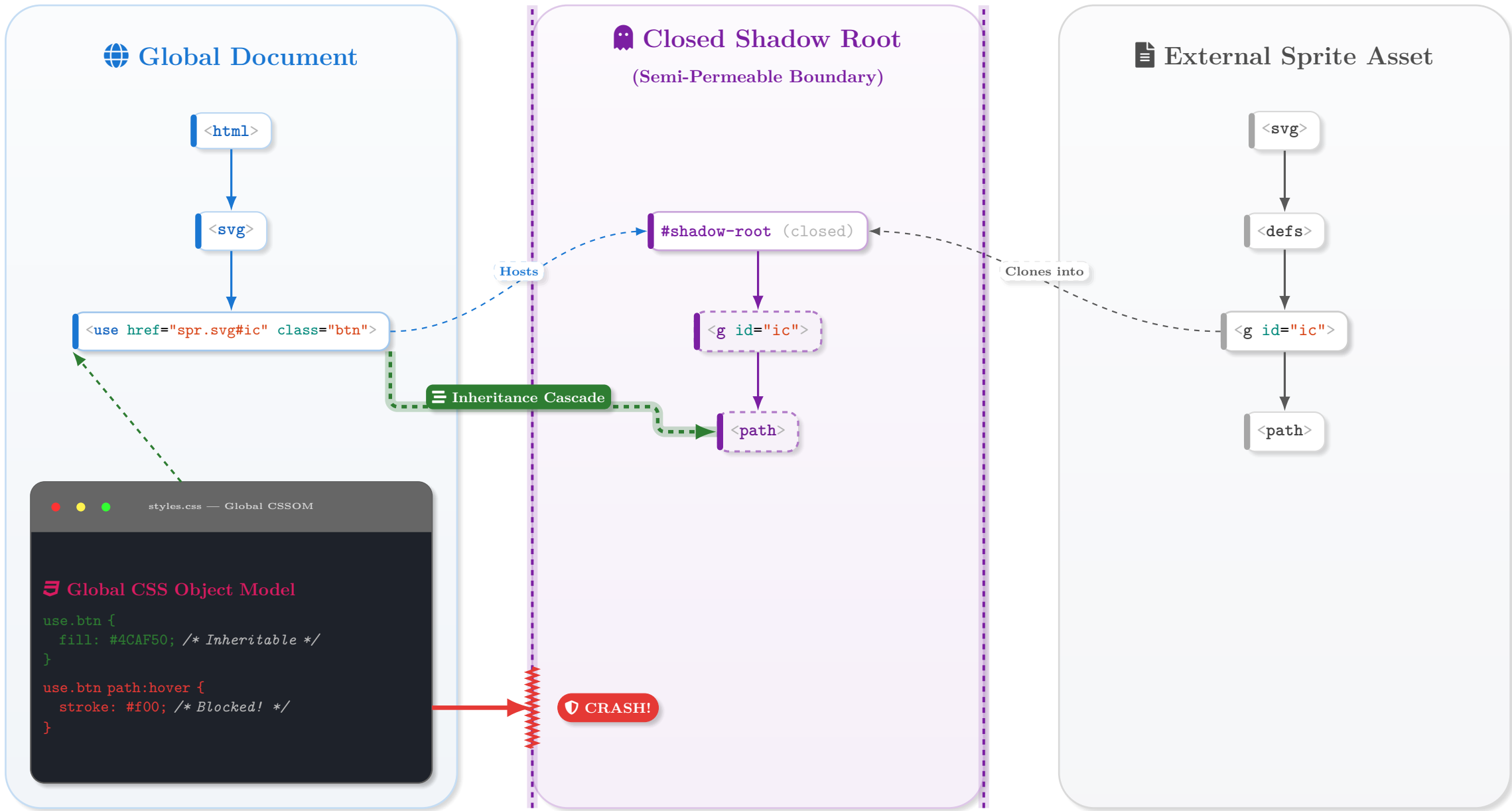


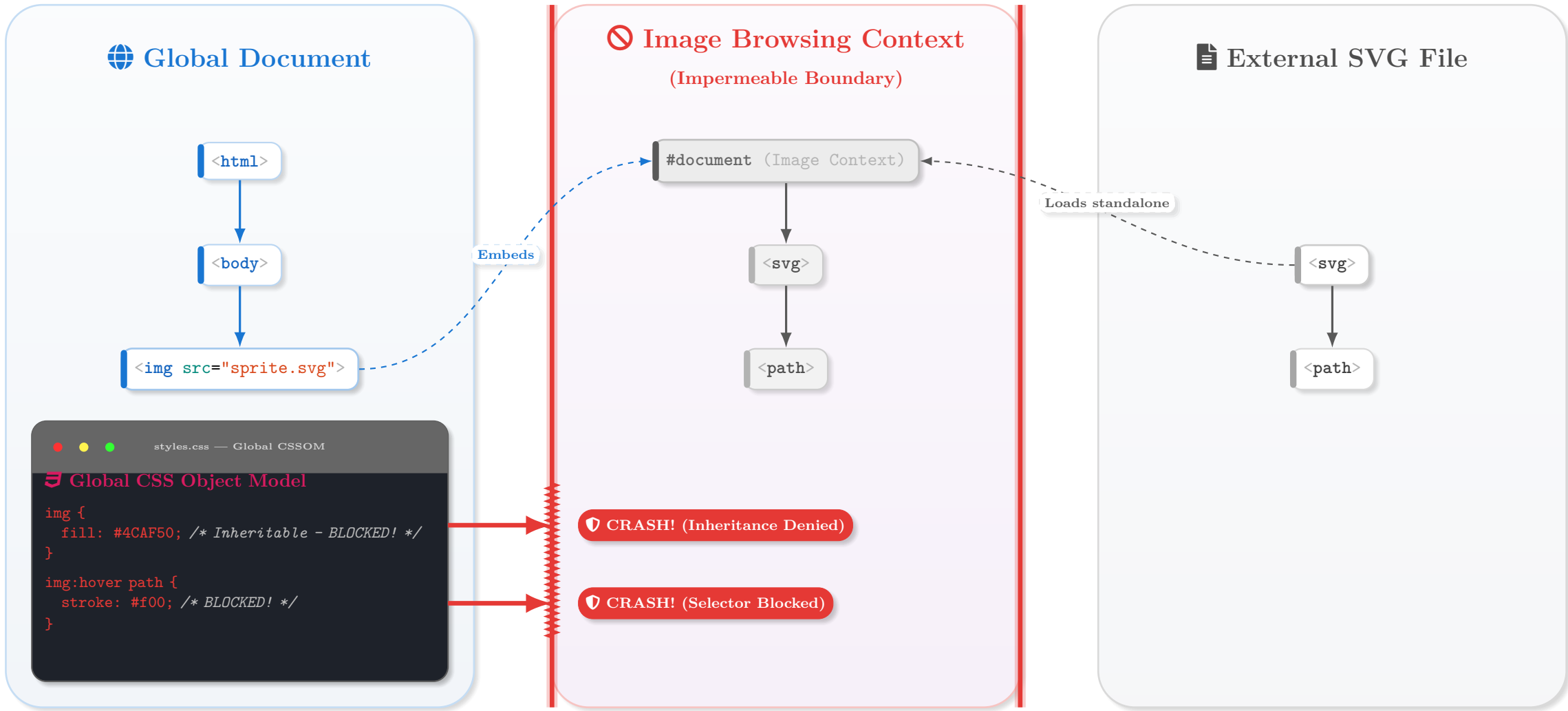
The Semi-Permeable Barrier: SVG <use>

Cross-Document Shadow Trees and Property Inheritance Loopholes



The Impermeable Barrier: SVG

Isolated Image Contexts and Total Encapsulation



1. Shadow Tree Instantiation

The `<use>` element instructs the browser to fetch an external SVG node and clone its entire tree structure into a **Closed Shadow Root** hosted directly by the `<use>` element.

2. The Permeable Inheritance Loophole

While selectors like `use path` are blocked by the Shadow boundary, *inheritable* CSS properties (`fill`, `color`, `stroke`) applied to the `<use>` tag will cascade downwards into the Ghost Nodes, enabling basic dynamic color theming.

3. The Strict Requirement

For the CSS inheritance loophole to work, the elements inside the external Sprite Asset (`<path>`) **must not** have hard-coded `fill` or `stroke` attributes, otherwise the hard-coded values will override the cascaded CSS.

4. The Impermeable Image Context

Importing an SVG via `<img>` or `background-image` does not create a permeable Shadow DOM. It creates a completely isolated Image Browsing Context. The host document's CSSOM cannot penetrate this boundary at all—even inheritable properties like `fill` are strictly blocked.